

You are required to complete one take-home assignment, which is a Coding Task designed to help us evaluate your coding style. Use this opportunity to showcase your understanding of coding best practices.

Please note that the assignment is confidential. Do not share or publish the questions, your code, design, or any related information with the public or anyone not involved in your hiring process with **VulcanLabs**

Coding & Implementation Task

Due to the ongoing pandemic, seating arrangements must follow specific distancing rules:

- People within the same group can sit together without any restrictions.
- Different groups must maintain a minimum distance from one another. This distance is measured using the *Manhattan distance* (diagonal moves are not allowed).

For example, in the seating layout below, the shortest distance between the two 1's is marked with x's, and the distance is 7. If `min_distance = 6`, this seating arrangement is acceptable.

```
[[1, X, X, X, X],  
 [0, 0, 0, 0, X],  
 [0, 0, 0, 0, X],  
 [0, 0, 0, 0, 1]]
```

We are looking for a solution to ensure cinema patrons follow social distancing rules. Your task is to develop a gRPC or Restful service that we can use for its cinema operations.

Requirements:

- The service should be configurable to support different cinema sizes and minimum distance rules, specified as: (rows, columns, `min_distance`).

- The service must provide a RPC method or API for querying a set of available seats that can be purchased together.
- Another RPC method or API should allow the reservation of specific seats, identified by their (row, column) coordinates ((0, 0) being the top left).
- Another RPC method or API should allow the cancellation of specific seats, identified by their (row, column) coordinates, with (0, 0) representing the top left corner.

Important Notes:

- Clearly state any assumptions you make regarding business requirements or preferences.
- Ensure that your solution highlights your coding style and technical knowledge.